A whole-circle limit set for a fixed backward-Euler map

Abstract

We study fixed-step backward Euler for the flat planar potential of Absil, Mahony and Andrews. Alaa and Pierre asked whether its near-boundary implicit orbits converge to a point or accumulate on the whole unit circle. We give explicit original initial points, independent of the step, for which the latter conclusion holds throughout a fixed interval of globally well-posed steps. The potential is unchanged. Its classical strict Lyapunov inequality gives square-summable increments and a vanishing gradient. An exact phase barrier for the unique implicit solve proves that the lifted angle tends to infinity with vanishing increments. The orbit therefore has the entire circle as its omega-limit set and infinite polygonal length. The potential, the general spiral phenomenon and earlier discrete phase arguments are prior; the result concerns the actual infinite orbit of this specified backward-Euler map.

Keywords: discrete dynamical system; Lyapunov function; implicit Euler; nonconvergent orbit; omega-limit set

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1 The energy and the original orbit

For $x \in \mathbb{R}^2$ with $0 < r = \|x\| < 1$, choose any polar angle θ and define

$$F(r,\theta) = \exp\left(-\frac{1}{1-r^2}\right)\left[1 - \frac{4r^4}{4r^4 + (1-r^2)^4}\sin\left(\theta - \frac{1}{1-r^2}\right)\right]. \tag{1}$$

Set $F(0) = e^{-1}$ and $F(x) = 0$ when $\|x\| \geq 1$. The angle in (1) is taken modulo 2π, so the function is well defined. This is the literal potential in Absil, Mahony and Andrews [1, equation (2.8)], repeated in Alaa and Pierre [2, Remark 2.7]. It is nonnegative and is not coercive.

Write $g = \nabla F$. We will verify directly that $F \in C^2(\mathbb{R}^2)$ and that its Hessian is bounded. Fix any

$$L \geq \max\left\{1, \sup_{x\in\mathbb{R}^2}\|\nabla^2 F(x)\|\right\}, \qquad 0 < h \leq \frac{1}{2L}. \tag{2}$$

The same positive step h is used at every index of the backward-Euler system

$$x_n = x_{n+1} + hg(x_{n+1}), \qquad n \geq 0. \tag{3}$$

Each equation will have a unique global solution. Thus no choice among implicit roots is part of the dynamics.

Theorem 1. *For every integer $N \geq 200$, start (3) at*

$$x_0 = \sqrt{1 - \frac{1}{N}}\big(\cos(N + \pi/2), \sin(N + \pi/2)\big). \tag{4}$$

For every fixed step satisfying (2), the resulting original orbit satisfies, for all $n \geq 0$,

$$\sqrt{1 - 1/100} < \|x_n\| < 1, \qquad 0 < F(x_{n+1}) < F(x_n), \tag{5}$$

$$F(x_n) - F(x_{n+1}) \geq h(1 - hL/2)\|g(x_{n+1})\|^2 \geq \frac{\|x_{n+1} - x_n\|^2}{2h}. \tag{6}$$

Identify the plane with $\mathbb{C}$. Define one real angle lift by

$$\theta_0 = N + \pi/2, \qquad \theta_{n+1} - \theta_n = \mathrm{Arg}(x_{n+1}/x_n), \tag{7}$$

where $\mathrm{Arg} \in (-\pi, \pi]$ is the principal argument. Every successive pair has positive scalar product, and this lift satisfies

$$\frac{\pi}{4} \leq \theta_n - \frac{1}{1 - \|x_n\|^2} \leq \frac{3\pi}{4}. \tag{8}$$

Moreover,

$$\|x_n\| \longrightarrow 1, \quad F(x_n) \longrightarrow 0, \quad g(x_n) \longrightarrow 0, \quad \theta_n \longrightarrow +\infty, \quad \theta_{n+1} - \theta_n \longrightarrow 0. \tag{9}$$

Consequently,

$$\omega((x_n)) = S^1, \qquad \sum_{n=0}^{\infty} \|x_{n+1} - x_n\|^2 \leq 2hF(x_0), \qquad \sum_{n=0}^{\infty} \|x_{n+1} - x_n\| = \infty. \tag{10}$$

Here $S^1 = \{x \in \mathbb{R}^2 : \|x\| = 1\}$ and $\omega((x_n))$ denotes the set of subsequential limits.

The starts in (4) work for the whole step interval (2). In particular, given $\varepsilon \in (0, 1/2)$, any integer

$$N > \max\{200, (2\varepsilon - \varepsilon^2)^{-1}\}$$

has $1 - \varepsilon < \|x_0\| < 1$. The theorem concerns this family of initial phases. It does not assert an open basin, all-phase behavior, a rate, or monotonicity of the angle.

Alaa and Pierre [2, Remark 2.7] discuss the same implicit scheme and potential. They record the Lyapunov and circle-containment setting and leave point convergence versus the whole-circle limit open. Theorem 1 gives the whole-circle alternative for the explicit starts (4); the uniform constants and quantifiers above are not attributed to that remark. The inequality (6) is the classical implicit-Euler energy inequality, also used in [2, Section 3.2]. The new orbit conclusion rests on (8), proved for each exact implicit solve.

The surrounding phenomenon has substantial precedent. The potential and a continuous spiral are due to [1]; its Section 3.2.1 also constructs a prescribed discrete descent sequence with a circular limit set. Besançon and Le [3, Appendix F] prove an actual

discrete spiral for a different, polynomial-amplitude and logarithmic-phase potential under explicit adaptive updates. Their energy, phase-trapping and lifted-angle arguments are relevant antecedents. We use elementary versions of those ideas, with exact geometry for the new-point gradient in (3). Section 5 states the scope of the inspected source comparison.

2 Well-posedness and energy confinement

For $0 < r < 1$, introduce

$$t = \frac{1}{1-r^2}, \qquad Q(t) = 4t^2(t-1)^2, \qquad a(t) = \frac{Q(t)}{1+Q(t)}, \qquad d(t) = \frac{1}{1+Q(t)}. \tag{11}$$

If $\phi = \theta - t$, then the unchanged formula (1) is

$$F(r,\theta) = e^{-t}(1 - a(t)\sin\phi), \qquad d = 1 - a. \tag{12}$$

Lemma 1. *The function F is globally C^2, with bounded compactly supported Hessian. Its gradient is globally L-Lipschitz for every L in (2). For any $x \in \mathbb{R}^2$ and $0 < h \le 1/(2L)$, the equation $x = y + hg(y)$ has exactly one solution. This solution is the unique global minimizer of*

$$y \longmapsto F(y) + \frac{\|y - x\|^2}{2h}. \tag{13}$$

Proof. Near the origin write $x = (u, v)$, $\rho = u^2 + v^2$ and $r = \sqrt{\rho}$. The angular part of F is exactly

$$-\frac{4r^3 e^{-1/(1-\rho)}}{4\rho^2 + (1-\rho)^4}\left[v\cos\frac{1}{1-\rho} - u\sin\frac{1}{1-\rho}\right].$$

The coefficients multiplying r^3u and r^3v are smooth functions of ρ near zero. Each of r^3u, r^3v is homogeneous of degree four away from zero. Its derivatives of orders $j \le 2$ are $O(r^{4-j})$ and extend with the required zero derivatives at the origin. This proves the asserted C^2 regularity there. The remaining radial term is smooth near zero.

Near the unit circle, every derivative of fixed order of the interior formula is bounded by $Ce^{-1/s}s^{-m}$ for some finite C, m, where $s = 1 - r^2 > 0$. The rational denominator stays away from zero, and local angular coordinates are smooth. These bounds tend to zero as $s \downarrow 0$, so the interior formula matches the zero extension with all derivatives. Away from the origin and circle the formula is smooth. Thus the Hessian is continuous and vanishes on and outside the circle; it is bounded and compactly supported. Integration on line segments proves the Lipschitz assertion.

Since $0 \le a < 1$, F is positive inside the disk and nonnegative everywhere. The function in (13) is coercive, and its Hessian is bounded below by

$$(1/h - L)I \succeq \frac{1}{2h}I.$$

It therefore has one global minimizer and no other critical point. Its critical equation is exactly $x = y + hg(y)$. ∎

Only global C^2 regularity is used. In particular, no smoothing of the literal origin is needed. To distinguish this from stronger smoothness wording, restriction to the horizontal axis gives

$$F(u,0) = e^{-1/(1-u^2)} + u|u|^3 H(u^2), \qquad H(0) = 4e^{-1}\sin 1 \neq 0,$$

with H smooth near zero. The one-sided fourth derivatives of the second term differ. The literal potential is therefore not globally C^4. The origin is explicitly distinguished in the regularity statement of [2, Remark 2.7].

Let $T_s(x)$ be the unique solution of $x = y + sg(y)$ for $0 < s \leq h$, and set $T_0(x) = x$. For $y = T_s(x)$, the lower Taylor inequality yields

$$F(x) - F(y) \geq s(1 - sL/2)\|g(y)\|^2 \geq \frac{\|x-y\|^2}{2s}. \tag{14}$$

If $\|x\| < 1$, then $\|T_s(x)\| < 1$: an endpoint on or outside the circle has zero gradient and would coincide with x.

Define $m(t) = e^{-t}d(t)$ for $t \geq 1$. Since

$$Q'(t) = 8t(t-1)(2t-1), \qquad a'(t) = \frac{Q'(t)}{(1+Q(t))^2} \geq 0, \tag{15}$$

we have

$$F(r,\theta) \geq m(t), \qquad m'(t) = -e^{-t}(d(t) + a'(t)) < 0. \tag{16}$$

The lower bound includes the origin with $t = 1$. At (4), $F(x_0) = m(N)$. Therefore (14) confines every original iterate, and every auxiliary point $y = T_s(x_n)$ for $0 \leq s \leq h$, to

$$0 < \|y\| < 1, \qquad F(y) \leq m(N), \qquad \frac{1}{1-\|y\|^2} \geq N \geq 200. \tag{17}$$

Indeed, the endpoint cannot be zero since $F(0) = m(1) > m(N)$, and (16) then proves the last inequality. This confinement is obtained before any phase argument and from the original initialization.

3 Exact phase barriers for an implicit step

We first record uniform elementary bounds on the larger annulus $t \geq 100$. Since $Q \geq t^4$ and $Q' \leq 16t^3$ for $t \geq 2$, equations (11) and (15) give

$$0 < d \leq t^{-4}, \qquad 0 < a' \leq 16t^{-5}, \qquad \frac{99}{100} \leq a < 1, \qquad a' \leq \frac{1}{100}. \tag{18}$$

Write $q = r^2 = 1 - t^{-1}$ and define

$$P(t,\phi) = -1 + (a - a')\sin\phi + a\cos\phi. \tag{19}$$

Differentiating (12) at fixed θ yields the exact force components

$$F_r = 2rt^2 e^{-t} P(t,\phi), \qquad F_\theta = -e^{-t} a \cos\phi. \tag{20}$$

In particular, $|P| < 4$. At the two phase boundaries,

$$P(t,\pi/4) = -1 + \sqrt{2}a - a'/\sqrt{2} \geq -1 + \frac{7}{5}\frac{99}{100} - \frac{1}{100} = \frac{47}{125} > 0, \tag{21}$$

$$P(t,3\pi/4) = -1 - a'/\sqrt{2} \leq -1. \tag{22}$$

Here $\sqrt{2} > 7/5$ follows by squaring positive numbers.

Lemma 2. *Suppose $\|x\| < 1$, $F(x) \leq m(N)$ and $N \geq 200$. Choose a real polar lift Θ of x such that*

$$\frac{\pi}{4} < \Theta - \frac{1}{1-\|x\|^2} < \frac{3\pi}{4}.$$

For every $s \in [0,h]$, let $y_s = T_s(x)$. Then $\langle x, y_s\rangle > 0$, and the lift

$$\theta(s) = \Theta - \operatorname{Arg}(x/y_s) \tag{23}$$

satisfies

$$\frac{\pi}{4} < \theta(s) - \frac{1}{1-\|y_s\|^2} < \frac{3\pi}{4}. \tag{24}$$

Proof. First, y_s depends continuously on s, including at zero. Indeed, $M = \sup\|g\|$ is finite by Lemma 1. Comparing the equations for y_s and y_u gives

$$\|y_s - y_u\| \leq sL\|y_s - y_u\| + |s-u|M, \qquad \|y_s - y_u\| \leq 2M|s-u|.$$

The confinement argument (17) applies to all these endpoints. Thus they are nonzero and have $t \geq N$. The parameter s is used only to verify one inverse map; the actual iteration still takes the fixed step h.

Let $y = y_s = re^{i\theta}$, $t = (1-r^2)^{-1}$ and $\phi = \theta - t$. Equations (20) and $x = y + sg(y)$ give

$$x = y(1 + A + ib), \qquad A = 2st^2 e^{-t} P(t,\phi), \qquad b = -\frac{se^{-t} a\cos\phi}{q}. \tag{25}$$

All quantities on the right are evaluated at the implicit endpoint y. Since $s \leq h \leq 1/2$, $q \geq 99/100 > 1/2$ and $e^t > t^4/24$, we obtain, uniformly in the phase,

$$|A| \leq 4t^2 e^{-t} < \frac{96}{t^2} \leq \frac{6}{625} < \frac{1}{4}, \qquad |b| \leq e^{-t} < \frac{24}{100^4} < \frac{1}{4}. \tag{26}$$

Thus $1 + A > 0$ and $\langle x, y\rangle = r^2(1+A) > 0$. The relative angle is exactly

$$\operatorname{Arg}(x/y) = \arctan\frac{b}{1+A} \in (-\pi/2, \pi/2). \tag{27}$$

It is continuous along the parameter interval, so (23) defines one continuous real lift with $\theta(0) = \Theta$.

Put $T = (1 - \|x\|^2)^{-1}$ and $\phi_x = \Theta - T$. Along this lift,

$$\phi_x - \phi = t - T + \operatorname{Arg}(1 + A + ib). \tag{28}$$

This identity is an equality of real numbers, not a congruence modulo 2π.

Suppose that at some $s > 0$ the phase reaches $\phi = \pi/4$. By (21) and (25), $A > 0$ and $b < 0$. Hence

$$\|x\|^2 = r^2\big((1 + A)^2 + b^2\big) > r^2, \qquad T > t, \qquad \operatorname{Arg}(1 + A + ib) < 0.$$

Equation (28) would imply $\phi_x < \pi/4$, a contradiction.

At the upper boundary $\phi = 3\pi/4$, we have $A < 0$, $b > 0$ and

$$|A| \geq 2st^2e^{-t}, \qquad b^2 = \frac{s^2e^{-2t}a^2}{2q^2} \leq 2s^2e^{-2t}.$$

In particular,

$$\frac{b^2}{|A|} \leq \frac{se^{-t}}{t^2} \leq \frac{e^{-t}}{2t^2} < \frac{12}{t^6} \leq \frac{12}{100^6} < \frac{1}{4}. \tag{29}$$

Together with (26), this proves

$$2A + A^2 + b^2 < -2|A| + \frac{1}{4}|A| + \frac{1}{4}|A| < 0.$$

Consequently $\|x\| < r$, $T < t$ and $\operatorname{Arg}(1 + A + ib) > 0$. Equation (28) would now imply $\phi_x > 3\pi/4$, again a contradiction. The tangential quadratic term b^2 has been retained in this argument.

The phase is continuous and begins strictly between the two boundaries at $s = 0$. Since it cannot reach either one for $s > 0$, it stays strictly between them throughout $[0, h]$. This proves (24). ∎

Apply Lemma 2 to $x = x_n$. The original phase is $\theta_0 - N = \pi/2$. At each endpoint $y_h = x_{n+1}$, equations (23) and (27) give

$$\theta(h) - \theta_n = -\operatorname{Arg}(x_n/x_{n+1}) = \operatorname{Arg}(x_{n+1}/x_n) \in (-\pi/2, \pi/2).$$

Thus the induction uses precisely the lift (7), with no subsequent phase reset. It proves (8), in fact with strict inequalities, and proves positivity of all successive scalar products.

4 Strict descent and the complete limit set

There are no critical points in the annulus $t \geq 100$. To see this, $F_\theta = 0$ in (20) forces $\cos\phi = 0$ since $a > 0$. If $\sin\phi = 1$, then

$$P = -d - a' < 0.$$

If $\sin\phi = -1$, then $P = -1 - a + a' < 0$ by (18). In either case $F_r \neq 0$. This exclusion does not require the phase strip.

Every actual endpoint lies in that annulus by (17). Applying (14) with $s = h$ therefore gives strict descent and all of (5)–(6). Telescoping proves the square-sum bound in (10). Since the originally chosen h is fixed and positive,

$$\|g(x_{n+1})\| = \frac{\|x_n - x_{n+1}\|}{h} \longrightarrow 0. \tag{30}$$

The orbit lies in the compact annulus $\sqrt{1 - 1/N} \leq \|x\| \leq 1$. Every cluster point is critical by continuity of g. The critical-point exclusion just proved implies that every cluster point lies on S^1. Compactness then gives $\|x_n\| \to 1$, and continuity gives $F(x_n) \to 0$.

Let $t_n = (1 - \|x_n\|^2)^{-1}$. The radial limit implies $t_n \to \infty$, so the real phase strip yields $\theta_n \to +\infty$. Applying (27) at the actual endpoint and using (26), we also obtain

$$|\theta_{n+1} - \theta_n| \leq \frac{4}{3}|b| \leq \frac{4}{3}e^{-t_{n+1}} \longrightarrow 0. \tag{31}$$

This establishes (9) without an angular-monotonicity assumption.

It remains to prove that no part of the circle is missed. Fix $\beta \in \mathbb{R}$. For sufficiently large integers k, let j_k be the first index such that

$$\theta_{j_k} \geq \beta + 2\pi k.$$

These indices exist, are positive, and tend to infinity. The first-crossing property and (31) give

$$0 \leq \theta_{j_k} - (\beta + 2\pi k) \leq \theta_{j_k} - \theta_{j_k - 1} \longrightarrow 0.$$

Thus $x_{j_k} \to (\cos\beta, \sin\beta)$, using the radial limit. Taking a subsequence of strictly increasing indices if necessary, every point of S^1 is a subsequential limit. The opposite containment was already proved, so $\omega((x_n)) = S^1$.

Finally, finite polygonal length would make (x_n) a Cauchy sequence in $\mathbb{R}^2$ and hence give a single limit. This contradicts the whole-circle limit set. The length is therefore infinite, and Theorem 1 is proved.

5 Relation to the earlier spiral examples

The conclusion is about the unchanged Cartesian recurrence (3). The auxiliary family T_s in Lemma 2 verifies a single unique inverse solve. It neither changes the orbit's fixed step nor approximates it by a continuous gradient trajectory. This distinction matters for a flat potential, where a finite-time comparison does not determine the infinite-time limit set.

The continuous spiral of [1] follows the phase relation $\theta = (1 - r^2)^{-1}$. Our initial phase is instead $\theta_0 - t_0 = \pi/2$, and the proof preserves a real interval around that value. The

separately prescribed discrete sequence in [1, Section 3.2.1] has angular samples $\theta_k = k\omega$ and radii $\sqrt{1 - 1/\theta_k}$, with ω/π irrational. Its limiting chord size is $2|\sin(\omega/2)| > 0$. It therefore cannot satisfy (3) with one fixed finite step while its gradient tends to zero. Those examples remain the antecedents of the potential and of the non-singleton limit-set phenomenon.

Truong and Nguyen [6, Example 3.3] test the same potential under explicit and backtracking gradient schemes. Their reported point convergence in that example is numerical and concerns different updates. Noll's graph-projection examples [4, Example 12.3] and [5, Examples 6–7] lead to the equation

$$x_{k-1} = x_k + f(x_k)\nabla f(x_k).$$

For $f = F$, this is associated with the objective $F^2/2$, rather than (3) for F. The inspected circling discussion is infinitesimal and does not supply the finite-step phase argument proved here.

The rigorous discrete spiral theorem of Besançon and Le [3, Appendix F] uses a different potential with polynomial amplitude and logarithmic phase, and an old-point adaptive update. Its descent, trapping and vanishing-increment mechanisms are closely related. Theorem 1 does not claim those general ideas or a new class of spiral potentials. It gives a quantitative all-index realization for the specific implicit scheme left open in [2, Remark 2.7], with the restricted initial family stated explicitly.

The source comparison uses the indexed final text of [2]; its final binary PDF was not recovered. The other cited comparisons use the inspected versions listed below. They do not constitute an exhaustive priority claim. None of these access qualifications is used as a mathematical premise of the self-contained proof.

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